Chapter 3 Vector Spaces

Section 3.6 Row Space and Column Space

Definition (Row space) If A is an $m \times n$ matrix, then the subspace of $\mathbf{R}^{1 \times n}$ spanned by the row vectors of A is called the **row space** of A.

Definition (Column space) If A is an $m \times n$ matrix, then the subspace of $\mathbf{R}^m$ spanned by the column vectors of A is called the **column space** of A.

Example Let $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.
The row space of A

$$\{a(1,1,0) + b(0,0,1) | a, b \in \mathbf{R}\} = \{(a,a,b) | a, b \in \mathbf{R}\}.$$

The column space of A is

$$\left\{ a\begin{pmatrix} 1 \\ 0 \end{pmatrix} + b\begin{pmatrix} 1 \\ 0 \end{pmatrix} + c\begin{pmatrix} 0 \\ 1 \end{pmatrix} \mid a, b, c \in \mathbf{R} \right\} = \mathbf{R}^2.$$

Theorem Two row equivalent matrices have the same row space.

Proof If B is row equivalent to A, then B can be formed from A by a finite sequence of row operations. Thus, the row vectors of B must be linear combinations of the row vectors of A. Consequently, the row space of B must be a subspace of the row space of A. Since A is row equivalent to B, by the same reasoning, the row space of A is a subspace of the row space of B.

Example Find a basis of the row space of $A = \begin{pmatrix} 1 & -2 & 3 \\ 2 & -5 & 1 \\ 1 & -4 & -7 \end{pmatrix}$.

Solution $A \longrightarrow \begin{pmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & 0 \end{pmatrix}$.

$\{(1, -2, 3), (0, 1, 5)\}$ is a basis for the row space of A.

Definition (Rank) The **rank of a matrix** A, denoted by $\text{rank}(A)$, is the dimension of the row space of A.

Example In the previous example, $\text{rank}(A) = 2$.

Definition in Sc3.2 (Null space) Let A be an $m \times n$ matrix. Let $N(A)$ denote the set of all solutions of the homogenous system $A\mathbf{x} = \mathbf{0}$. That is,

$$N(A) = \{\mathbf{x} \in \mathbf{R}^n | A\mathbf{x} = \mathbf{0}\}$$

where $N(A)$ is a subspace of $\mathbf{R}^n$, and is called the **null space** of A.

Definition (Nullity) The dimension of the null space of a matrix is called the **nullity** of the matrix.

Example Let $A = \begin{pmatrix} 1 & -2 & 1 & 1 & 2 \\ -1 & 3 & 0 & 2 & -2 \\ 0 & 1 & 1 & 3 & 4 \\ 1 & 2 & 5 & 13 & 5 \end{pmatrix}$. Find the nullity of A.

Solution The reduced row echelon form of A is $\begin{pmatrix} 1 & 0 & 3 & 7 & 0 \\ 0 & 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$.

$$\begin{aligned} N(A) &= \{\mathbf{x} = (x_1, x_2, x_3, x_4, x_5)^T : A\mathbf{x} = \mathbf{0}\} \\ &= \left\{ (x_1, x_2, x_3, x_4, x_5)^T : \begin{pmatrix} 1 & 0 & 3 & 7 & 0 \\ 0 & 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \right\} \\ &= \left\{ \begin{pmatrix} -3\alpha - 7\beta \\ -\alpha - 3\beta \\ \alpha \\ \beta \\ 0 \end{pmatrix} \right\} = \left\{ \alpha \begin{pmatrix} -3 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} -7 \\ -3 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\} \end{aligned}$$

$\left\{ \begin{pmatrix} -3 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -7 \\ -3 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}$ is a basis of $N(A)$. The nullity of A is 2.

Theorem (The Rank-Nullity Theorem) If A is an $m \times n$ matrix, then the rank of A plus the nullity of A equals n.

Proof Let U be the reduced row echelon form of A. The system $A\mathbf{x} = \mathbf{0}$ is equivalent to the system $U\mathbf{x} = \mathbf{0}$. If $\text{rank}(A) = r$, then U will have r nonzero rows, and consequently the system $U\mathbf{x} = \mathbf{0}$ will involve r lead variables and $n - r$ free variables. The dimension of $N(A)$ will equal the number of free variables, i.e; $\text{rank}(A) + \dim N(A) = n$.

Theorem (Consistency Theorem for Linear Systems) A linear system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ is in the column space of A.

Proof Let $\mathbf{a}_i$ be the ith column of A, and let $\mathbf{x} = (x_1, x_2, \cdots, x_n)^T$. Note that $A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \ + x_n\mathbf{a}_n$. Thus the system $A\mathbf{x} = \mathbf{b}$ is consistent, if and only if there exists $\mathbf{x}' = (x_1', x_2', \cdots, x_n')^T$ such that $\mathbf{b} = x_1'\mathbf{a}_1 + x_2'\mathbf{a}_2 + \ + x_n'\mathbf{a}_n$, if and only if $\mathbf{b}$ is in the column space of A.

Theorem

1. The linear system $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b} \in \mathbf{R}^n$ if and only if the column vectors of A span $\mathbf{R}^n$.
2. The system $A\mathbf{x} = \mathbf{b}$ has at most one solution for every $\mathbf{b} \in \mathbf{R}^n$ if and only if the column vectors of A are linearly independent.
3. The system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b} \in \mathbf{R}^n$ if and only if the column vectors of A form a basis of $\mathbf{R}^n$.

Proof of (1) By the theorem of the previous slide, the system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ is in the column space of A. It follows that $A\mathbf{x} = \mathbf{b}$ will be consistent for every $\mathbf{b} \in \mathbf{R}^n$ if and only if the column vectors of A span $\mathbf{R}^n$.

Ex $A = \begin{bmatrix} 1 & 2 & -1 & 1 \\ 2 & 4 & -3 & 0 \\ 1 & 2 & 1 & 5 \end{bmatrix}$

$\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -3 \\ 1 \end{bmatrix}$

-3+2

(a) $\begin{bmatrix} 1 & 2 & -1 & 1 \\ 2 & 4 & -3 & 0 \\ 1 & 2 & 1 & 5 \end{bmatrix}$ column basis

$\xrightarrow[R_3 - R_1 \to R_1]{R_2 - 2R_1 \to R_2} \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 2 & 4 \end{bmatrix}$

$\xrightarrow[2 \times (-1) \to R_2]{R_3 + 2R_2 \to R_3} \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\xrightarrow{R_1 + R_2 \to R_1} \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\alpha \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -3 \\ 0 \\ -2 \\ 1 \end{bmatrix} = 0$

$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{matrix} -2\alpha - 3\beta \\ \alpha \\ -2\beta \\ \beta \end{matrix}$

$\text{null}(A) = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$

(1)

$\text{row} \{(1, 2, 0, 3), (0, 0, 1, 2)\}$

Proof of (2) If $A\mathbf{x} = \mathbf{b}$ has at most one solution for every $\mathbf{b}$, then, in particular, the system $A\mathbf{x} = \mathbf{0}$ can have only the trivial solution, and hence the column vectors of A must be linearly independent.
Conversely, if the column vectors of A are linearly independent, $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. Now, if $\mathbf{x}_1$ and $\mathbf{x}_2$ were both solutions of $A\mathbf{x} = \mathbf{b}$, then $\mathbf{x}_1 - \mathbf{x}_2$ would be a solution of $A\mathbf{x} = \mathbf{0}$,

$$A(\mathbf{x}_1 - \mathbf{x}_2) = A\mathbf{x}_1 - A\mathbf{x}_2 = \mathbf{b} - \mathbf{b} = \mathbf{0}.$$

It follows that $\mathbf{x}_1 - \mathbf{x}_2 = \mathbf{0}$, and hence $\mathbf{x}_1$ must equal $\mathbf{x}_2$.

Proof of (3) It follows from (1), (2) and the definition of a basis.

Theorem For any matrix A, the dimension of the row space of A equals the dimension of the column space of A.

Proof If A is an $m \times n$ matrix of rank r, the row echelon form U of A will have r leading 1's. The columns of U corresponding to the lead 1's will be linearly independent.

Let U_L denote the matrix obtained from U by deleting all the columns corresponding to the free variables. Delete the same columns from A and denote the new matrix by A_L. The matrices A_L and U_L are row equivalent. Thus, if $\mathbf{x}$ is a solution of $A_L\mathbf{x} = \mathbf{0}$, then x must also be a solution of $U_L\mathbf{x} = \mathbf{0}$. Since the columns of U_L are linearly independent, $\mathbf{x}$ must equal $\mathbf{0}$. From the previous theorem, the columns of A_L are linearly independent.

Proof (continuity) Since A_L has r columns, the dimension of the column space of A is at least r. We have proved that, for any matrix, the dimension of the column space is greater than or equal to the dimension of the row space. Applying this result to the matrix A^T , we see that

$$\begin{aligned}\dim(\text{row space of } A) &= \dim(\text{column space of } A^T) \\ &\geq \dim(\text{row space of } A^T) \\ &= \dim(\text{column space of } A)\end{aligned}$$

Thus, for any matrix A, the dimension of the row space must equal the dimension of the column space.

Example Let $A = \begin{pmatrix} 2 & -4 & 3 & 0 & 1 & 6 \\ 1 & -2 & -2 & 14 & -4 & 15 \\ 1 & -2 & 1 & 2 & 1 & -1 \\ -2 & 4 & 0 & -12 & 1 & -7 \end{pmatrix}$. Find a basis of column space of a matrix A.

Solution $A \longrightarrow \begin{pmatrix} 1 & -2 & 0 & 6 & 0 & 1 \\ 0 & 0 & 1 & -4 & 0 & 3 \\ 0 & 0 & 0 & 0 & 1 & -5 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$

$\{(2,1,1,-2)^T, (3,-2,1,0)^T, (1,-4,1,1)^T\}$ is a basis of the column space of A.

Theorem (Equivalent Conditions for Nonsingularity)

Let A be an $n \times n$ matrix. The following are equivalent:

- A is nonsingular.
- $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{0}$.
- A is row equivalent to I. (I is the reduced row echelon form of A. A can be written as a product of elementary matrices.)
- The system $A\mathbf{x} = \mathbf{b}$ has exactly one solution for every $\mathbf{b} \in \mathbf{R}^m$, (which is $\mathbf{x} = A^{-1}\mathbf{b}$).
- $\det A \neq 0$.
- Columns of A form a basis of $\mathbf{R}^n$.
- The rank of A is n.
- Rows of A form a basis of $\mathbf{R}^{1\times n}$.
- The nullity of A is 0.

Chapter 3 Vector Spaces

Section 3.6 Row Space and Column Space (Extra)

Theorem Let A be an $m \times n$ matrix.

1. $\text{rank}(A) + \dim N(A) = n$.
2. $\text{rank}(A) = \text{rank}(A^T)$ [The dimension of column space is the rank].
3. $0 \leq \text{rank}(A) \leq min(n, m)$.
4. $\text{rank}(A) = 0$ if and only if $A = O$.

n-k (numbers of free variables)

(Rank-Nullity Thm) 重点

k (number of leading ones)

leading ones + free variables = dimension

Thm (Equivalent statement for nonsingularity)

① $A: n \times n$
② A: nonsingular
③ $Ax = 0$ (only trivial sol $x = 0$)
④ $A \xrightarrow{\text{rref}} I$
⑤ $Ax = b$ (unique sol for any b)
⑥ $\det(A) \neq 0$
⑦ $\text{rank}(A) = n$ [$A \xrightarrow{\text{rref}} I$]
⑧ Columns of A form a basis for $\mathbb{R}^n$
⑨ Rows of A form a basis for $\mathbb{R}^{1 \times n}$

Thm ①. Rank-Nullity Theorem:

$A_{m\times n}$ $\quad \underline{\text{rank}(A)} + \dim(N(A)) = n$

Pf: $E_k \cdots E_1 A = \underline{U}$ upper triangular matrix (leading ones)

$A\vec{x} = 0 \Rightarrow U\vec{x} = 0$

subspace

Theorem Let A be an $m \times n$ matrix.

1. $\text{rank}(A) + \dim N(A) = n$.
2. $\text{rank}(A) = \text{rank}(A^T)$ [The dimension of column space is the rank].
3. $0 \leq \text{rank}(A) \leq \min(n, m)$.
4. $\text{rank}(A) = 0$ if and only if $A = O$.

number of leading ones

We are going to show

Theorem Let A be an $m \times n$ matrix.

5. $\text{rank}(A) = \text{rank}(A^T A) = \text{rank}(AA^T)$.
6. If P is $m \times m$, Q is $n \times n$, and both P and Q are nonsingular, then $\text{rank}(PAQ) = \text{rank}(A)$.
7. For any $m \times n$ matrix B, $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$.
8. For any $n \times r$ matrix B, $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$.

If Q nonsingular
$\text{rank}(AQ) = \text{rank}(A)$

P nonsingular, $\text{rank}(PA) = \text{rank}(A)$ ($P: m \times m$, $A\ m \times n$, $PA: m \times n$)

Claim: $N(A) = N(PA)$

Pf: Let $x \in N(A)$. then $Ax = 0 \Rightarrow (PA)x = P(Ax) = 0$

$x \in N(A)$. $Ax = 0$

$\Rightarrow x \in N(PA)$

$\Rightarrow N(A) \subset N(PA)$ ①

Let $x \in N(PA)$

$\Rightarrow (PA)x = 0 = P(Ax) = 0$ (P nonsingular) $\Rightarrow Ax = 0$ (only trivial sol) $\Rightarrow x \in N(A)$

$\Rightarrow N(PA) \subset N(A)$ ②

From ① and ② $\Rightarrow N(A) = N(PA)$

$\text{null}(A) = \text{null}(PA) \Rightarrow$

$\text{rank}(A) = n - \text{null}(A)$

$\text{rank}(PA) = n - \text{null}(PA)$

$\Rightarrow \text{rank}(A) = \text{rank}(PA)$

Thm 5 : $\text{rank}(A) = \text{rank}(AA^T) = \text{rank}(A^TA)$

Theorem (5) Let A be an $m \times n$ matrix. $\text{rank}(A) = \text{rank}(A^T A) = \text{rank}(AA^T)$.

Example

Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$.

Then $A^T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$, $A^T A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $AA^T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

$\text{rank}(A) = 2$, $\text{rank}(A^T A) = 2$, $\text{rank}(AA^T) = 2$.

Theorem (6) Let A be an $m \times n$ matrix. If P is $m \times m$, Q is $n \times n$, and both P and Q are nonsingular, then $\text{rank}(PAQ) = \text{rank}(A)$.

Example Let $A = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \end{pmatrix}$, $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $Q = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$,

$R = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

Note that P, Q are nonsingular because $\det(P) = -1 = \det(Q)$. But R is singular because $\det(R) = 0$.

Then $PAI = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}$, $IAQ = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}$, $IAR = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 5 & 0 \end{pmatrix}$.

$\text{rank}(A) = 2$, $\text{rank}(PAI) = \text{rank}(A) = 2$, $\text{rank}(IAQ) = \text{rank}(A) = 2$, $\text{rank}(IAR) = 1 \neq \text{rank}(A)$.

Theorem (5) Let A be a matrix. $\text{rank}(A) = \text{rank}(A^TA) = \text{rank}(AA^T)$.

Proof of $\text{rank}(A) = \text{rank}(A^TA)$

Let $\mathbf{x} \in N(A)$. Then $A\mathbf{x} = \mathbf{0}$ and $(A^TA)\mathbf{x} = A^T(A\mathbf{x}) = A^T\mathbf{0} = \mathbf{0}$. So $\mathbf{x} \in N(A^TA)$. Hence $N(A)$ is a subspace in $N(A^TA)$.

Let $\mathbf{y} \in N(A^TA)$ and $\mathbf{z} = A\mathbf{y}$. Then $(A^TA)\mathbf{y} = \mathbf{0}$ and

$$\mathbf{z}^T\mathbf{z} = (A\mathbf{y})^T(A\mathbf{y}) = (\mathbf{y}^TA^T)A\mathbf{y} = \mathbf{y}^T(A^TA\mathbf{y}) = \mathbf{y}^T\mathbf{0} = \mathbf{0}.$$

Note the $\mathbf{y}^T\mathbf{0}$ is the 1×1 matrix $\mathbf{0}$. Since $\mathbf{z}$ is a column vector such that $\mathbf{z}^T\mathbf{z} = \mathbf{0}$, we have $A\mathbf{y} = \mathbf{z} = \mathbf{0}$ (why?). So $\mathbf{y} \in N(A)$, and $N(A^TA)$ is a subspace of $N(A)$.

So $N(A^TA) = N(A)$. By the Rank-Nullity Theorem,
$\text{rank}(A^TA) = n - \dim(N(A^TA)) = n - \dim(N(A)) = \text{rank}(A)$.

Let $x \in N(A)$

Theorem (Half of 6) Let A be an $m \times n$ matrix, P be $m \times m$ nonsingular matrix. Then $\text{rank}(PA) = \text{rank}(A)$.

$P(Ax) = 0$

$(PA)x =$

$\Rightarrow x \in N(PA)$

Proof

Let $\mathbf{x} \in N(A)$. Then $PA\mathbf{x} = P\mathbf{0} = \mathbf{0}$ and hence $\mathbf{x} \in N(PA)$. Thus $N(A)$ is a subspace of $N(PA)$.

On the other hand, if $\mathbf{y} \in N(PA)$, then $P(A\mathbf{y}) = PA\mathbf{y} = \mathbf{0}$ and hence $A\mathbf{y} \in N(P)$. But $N(P) = \{\mathbf{0}\}$ since P is nonsingular. Therefore $A\mathbf{y} = \mathbf{0}$ and hence $\mathbf{y} \in N(A)$. Thus $N(PA)$ is a subspace of $N(A)$.

Now, $N(PA) = N(A)$. It follows from the Rank-Nullity Theorem that $\text{rank}(A) = n - dim N(A) = n - dim N(PA) = \text{rank}(PA)$.

Theorem (7) Let A be an $m \times n$ matrix. Let B be an $n \times r$ matrix. Then $\text{rank}(AB) \leq min\{\text{rank}(A), \text{rank}(B)\}$.

Example Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}$. $AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$

$\text{rank}(A) = 2$, $\text{rank}(B) = 2$, $\text{rank}(AB) = 1$.

Example Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$. $AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$

$\text{rank}(A) = 1$, $\text{rank}(B) = 2$, $\text{rank}(AB) = 1$.

Example Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$. $AB = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$

$\text{rank}(A) = 2$, $\text{rank}(B) = 2$, $\text{rank}(AB) = 2$.

Theorem (8) Let A, B be $m \times n$ matrices. Then $\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$.

Example Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$. $A+B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$

$\text{rank}(A) = 2$, $\text{rank}(B) = 2$, $\text{rank}(A+B) = 2$.

Example Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. $A+B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$\text{rank}(A) = 2$, $\text{rank}(B) = 1$, $\text{rank}(A+B) = 3$.

Theorem (7) Let A be an $m \times n$ matrix. Let B be an $n \times r$ matrix. Then $\text{rank}(AB) \leq min\{\text{rank}(A), \text{rank}(B)\}$.

Proof

Let $Col(A)$, $Col(AB)$ be the column space of A and AB respectively.

Let $\mathbf{d} \in Col(AB)$. By Theorem $(*)$, $\mathbf{d} = AB\mathbf{x}$ for some $\mathbf{x} \in \mathbf{R}^r$. Let $\mathbf{y} = B\mathbf{x}$. Since $\mathbf{d} = A\mathbf{y}$, it follows that $\mathbf{d}$ is in $Col(A)$ by Theorem $(*)$. Hence $Col(AB)$ is a subspace of $Col(A)$. Hence
$\text{rank}(AB) = \dim(Col(AB)) \leq \dim(Col(A)) = \text{rank}(A)$.

$$
\begin{aligned}
\text{rank}(AB) &= \text{rank}((AB)^T) && \text{dim. of col. sp.=rank} \\
&= \text{rank}(B^T A^T) \\
&\leq \text{rank}(B^T) && \text{we have just shown } \text{rank}(CD) \leq \text{rank}(C) \\
&= \text{rank}(B) && \text{dim. of col. sp.=rank}
\end{aligned}
$$

Theorem (81) Let A, B be $m \times n$ matrices. Then $\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$.

Proof Let $Col(A) + Col(B) = \{\mathbf{x}+\mathbf{y} | \mathbf{x} \in Col(A), \mathbf{y} \in Col(B)\}$.
Let $\mathbf{d} \in Col(A+B)$. Then $(A+B)\mathbf{x} = \mathbf{d}$ for some $\mathbf{x}$. Then $\mathbf{d} = A\mathbf{x} + B\mathbf{x}$.
Since $A\mathbf{x}$ is in $Col(A)$ and $B\mathbf{x}$ is in $Col(B)$, we have $\mathbf{d} \in Col(A) + Col(B)$. So $Col(A+B)$ is a subspace of $Col(A) + Col(B)$.

Hence,

$$\begin{aligned}
&\text{rank}(A+B) \\
&= \dim(Col(A+B)) \\
&\leq \dim(Col(A) + Col(B)) \\
&= \dim(Col(A)) + \dim(Col(B)) - \dim(Col(A) \cap Col(B)) \\
&\leq \dim(Col(A)) + \dim(Col(B)) \\
&= \text{rank}(A) + \text{rank}(B)
\end{aligned}$$

by the dimension theorem at the end of the slides

$C = A+B = (\vec{c}_1 \cdots \vec{c}_n)$

$\mathbf{d} = x_1\vec{c}_1 + \cdots + x_n\vec{c}_n = (\vec{c}_1 \cdots \vec{c}_n)\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (A+B)\vec{x}$

Theorem (8) Let A, B be $m \times n$ matrices. Then $\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$.

Proof 2 Let $Q = \left(\begin{array}{c|c} A & O \\ \hline O & B \end{array}\right)$. Then $\text{rank}(Q) = \text{rank}(A) + \text{rank}(B)$. (You can see this by converting Q into row-echelon form).

Since Q and $Q' = \left(\begin{array}{c|c} A & O \\ \hline A & B \end{array}\right)$ are row equivalent, $\text{rank}(Q) = \text{rank}(Q')$.

Since $Q'^T = \left(\begin{array}{c|c} A^T & A^T \\ \hline O & B^T \end{array}\right)$ and $Q'' = \left(\begin{array}{c|c} A^T & A^T \\ \hline A^T & A^T + B^T \end{array}\right)$ are row equivalent, $\text{rank}(Q'^T) = \text{rank}(Q'')$.

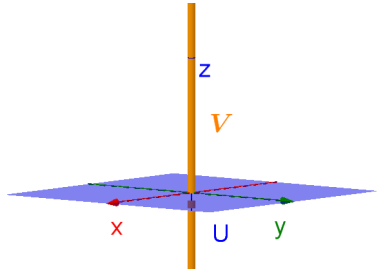
Since dimension of the column space of a matrix equals the rank, $\text{rank}(Q) = \text{rank}(Q') = \text{rank}(Q'^T) = \text{rank}(Q'') = \text{rank}(Q''^T)$.

Note that Q''^T has a submatrix $A + B$. Since the rank of a submatrix cannot exceed the rank of the whole matrix (why?), $\text{rank}(A+B) \leq \text{rank}(Q) = \text{rank}(A) + \text{rank}(B)$.

Dimension Theorem Let U, V be subspaces of a vector space W. Let $U + V = \{\mathbf{u} + \mathbf{v} | \mathbf{u} \in U, \mathbf{v} \in V\}$. Then
$\dim(U) + \dim(V) = \dim(U + V) + \dim(U \cap V)$.

Example

$U = \{(x, y, 0)^T | x, y \in \mathbf{R}\}$
$V = \{(0, 0, z)^T | z \in \mathbf{R}\}$
$U \cap V = \{\mathbf{0}\}$,
$U + V = \mathbf{R}^3$
$\dim U = 2$
$\dim V = 1$
$\dim(U \cap V) = 0$
$\dim(U + V) = 3$



Dimension Theorem Let U, V be subspaces of a vector space W.
Let $U + V = \{\mathbf{u} + \mathbf{v} | \mathbf{u} \in U, \mathbf{v} \in V\}$. Then
$\dim(U) + \dim(V) = \dim(U + V) + \dim(U \cap V)$.

Example
$U = \{(0, y, 0)^T | y \in \mathbf{R}\}$
$V = \{(0, 0, z)^T | z \in \mathbf{R}\}$
$U \cap V = \{\mathbf{0}\}$, $\quad U + V = \{(0, y, z)^T | y, z \in \mathbf{R}\}$
$\dim U = 1$, $\dim V = 1$
$\dim(U \cap V) = 0$, $\dim(U + V) = 2$

Let U, V be subspace of W

$U \cap V$ is also a subspace of W

check: ① not empty $\Rightarrow 0 \in U$ (every subspace must contain zero vector)

$0 \in V$

$\Rightarrow 0 \in U \cap V$

② Closure: Addition $\Rightarrow$ let $x \in U \cap V,\ y \in U \cap V$

$x \in U,\ y \in U \Rightarrow x+y \in U$

$x \in V,\ y \in V \Rightarrow x+y \in V$

$\Rightarrow x+y \in U \cap V$

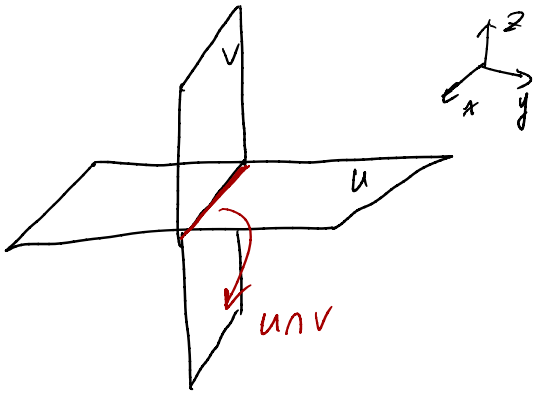
③ Closure: Multiplication $\Rightarrow$ Let $x \in U \cap V,\ \alpha \in \mathbb{R}$

(Scalar)

$x \in U \Rightarrow \alpha x \in U$

$x \in V \Rightarrow \alpha x \in V$

$\Rightarrow \alpha x \in U \cap V$



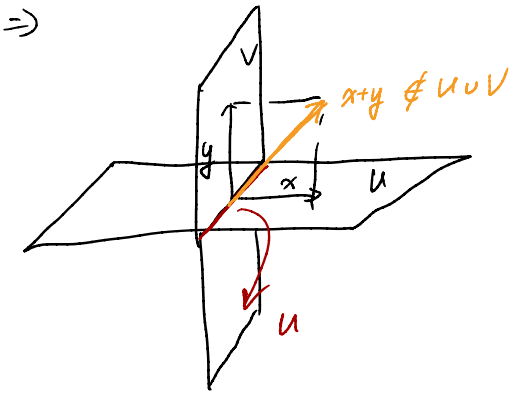
$U = \{x, y, 0\},\ x, y \in \mathbb{R}\}$

$V = \{(x, 0, z),\ x, z \in \mathbb{R}\}$

$U \cap V = \{(x, 0, 0),\ x \in \mathbb{R}\}$

* $U \cup V$ is not a subspace in general.

Pf of (*): ① Closure: Addition $\Rightarrow$



$U + V$ is a subspace of W

$\{u+v \mid u+v :\ u \in U,\ v \in V\}$

Pf: ① $0 \in U,\ 0 \in V,\ 0 \in U+V \Rightarrow U+V \neq \varnothing$

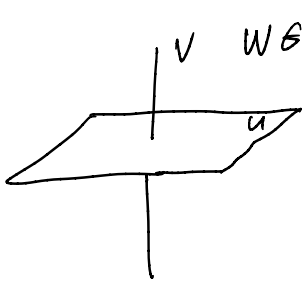
② Let $x \in U+V,\ y \in U+V$

$x = u_1 + v_1,\ y = u_2 + v_2,\ u_1, u_2 \in U,\ v_1, v_2 \in V$

$\Rightarrow x+y = (u_1+u_2) + (v_1+v_2) \in U+V$

③ Let $x = u+v,\ u \in U,\ v \in V,\ \alpha \in \mathbb{R}$

$\alpha x = \alpha u + \alpha v \in U+V$



$U+V \in \mathbb{R}^3$ (**)

$U = \{(x, y, 0),\ x \in \mathbb{R},\ y \in \mathbb{R}\}$

$V = \{(0, 0, z),\ z \in \mathbb{R}\}$

Pf (**) Let $W \in \mathbb{R}^3$

1st sol: $W = (x, y, z) = (x, y, 0) + (0, 0, z) \in U+V$

$\Rightarrow \mathbb{R}^3 \subset U+V$

$U+V \in \mathbb{R}^3$

$\Rightarrow U+V = \mathbb{R}^3$

2nd: Let $W = (x, y, z) \in \mathbb{R}^3$

$W = (x, y, 0) + (0, 0, z) \in U+V$

or $W = (0, y, 0) + (x, 0, z) \in U+V$

Thm: Let U, V be subspace of a vector space W.

Then $\dim(U) + \dim(V) = \dim(U+V) + \dim(U \cap V)$

↳ find the basis

Pf: Let $\{x_1, \dots, x_k\}$ be a basis for $U \cap V$ (extend)

$\dim(U \cap V) = k$.

is a basis for U

since $U \cap V$ is a subspace of U. $\Rightarrow \{x_1, \dots, x_k, y_1, \dots y_m\}$

$\{x_1, \dots x_k, z_1, \dots z_n\}$ for V

Claim: $\{x_1, \dots x_k, y_1, \dots y_m, z_1, \dots, z_n\}$ for $U+V$

pf of Claim: (1) Let $w \in U+V$

$\Rightarrow w = u+v,\ u \in U,\ v \in V$

then $u = (a_1 x_1 + \cdots a_k x_k) + (b_1 y_1 + \cdots + b_m y_m)$

$v = (c_1 x_1 + \cdots + c_k x_k) + (d_1 z_1 + \cdots + d_n z_n)$

for some a_i, b_i, c_i, d_i

Then $u+v = (a_1+c_1)x_1 + \cdots + (a_k+c_k)x_k + (b_1 y_1 + \cdots + b_m y_m) + (d_1 z_1 + \cdots + d_n z_n)$

So $U+V = \text{span}\{x_1, \cdots x_k, y_1, \cdots, y_m, z_1, \cdots z_n\}$

(2) Let $a_1 x_1 + \cdots a_k x_k + b_1 y_1 + \cdots + b_m y_m + c_1 z_1 + \cdots + c_n z_n = 0$

(Show it is linearly indepts)

$\Rightarrow \underbrace{a_1 x_1 + \cdots + a_k x_k + b_1 y_1 + \cdots + b_m y_m}_{\in U} = \underbrace{-c_1 z_1 - \cdots - c_n z_n}_{\in V}$

both LHS & RHS belongs to $U \cap V$

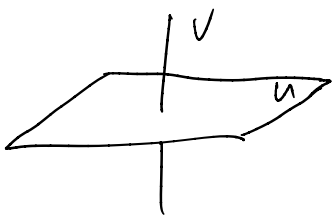
$c_1 z_1 + \cdots + c_n z_n \in U \cap V$

$\Rightarrow c_1 z_1 + \cdots + c_n z_n = d_1 x_1 + \cdots + d_k x_k$

$= 0$

$\Rightarrow c_1 = \cdots = c_n = d_1 = \cdots = d_k = 0$

★ 0 subspace has no basis



$U = V^{\perp}$

$V = U^{\perp}$

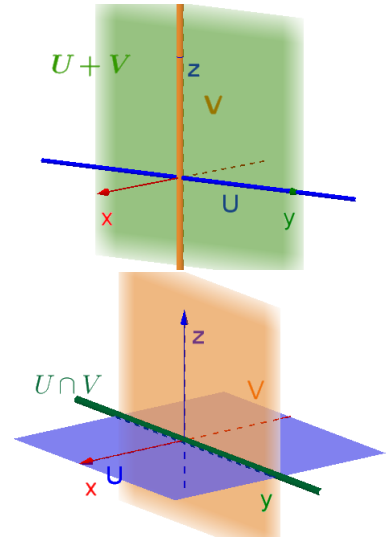
orthogonal 正交

orthogonal complement

Dimension Theorem Let U, V be subspaces of a vector space W.
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Example
$U = \{(0, y, 0)^T | y \in \mathbf{R}\}$
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$\dim U = 1$, $\dim V = 1$
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Example
$U = \{(x, y, 0)^T | x, y \in \mathbf{R}\}$
$V = \{(0, y, z)^T | y, z \in \mathbf{R}\}$
$U \cap V = \{(0, y, 0)^T | y, 0 \in \mathbf{R}\}$, $U + V = \mathbf{R}^3$
$\dim U = 2$, $\dim V = 2$
$\dim(U \cap V) = 1$, $\dim(U + V) = 3$



Dimension Theorem Let U, V be subspaces of a vector space W. Let $U + V = \{\mathbf{u} + \mathbf{v} | \mathbf{u} \in U, \mathbf{v} \in V\}$. Then $\dim(U) + \dim(V) = \dim(U + V) + \dim(U \cap V)$.

Idea of a proof Let $\{\mathbf{x}_1, \cdots, \mathbf{x}_k\}$ be a basis for $U \cap V$. Then we can extend $\{\mathbf{x}_1, \cdots, \mathbf{x}_k\}$ to a basis $\{\mathbf{x}_1, \cdots, \mathbf{x}_k, \mathbf{y}_1, \cdots, \mathbf{y}_\ell\}$ of U and $\{\mathbf{x}_1, \cdots, \mathbf{x}_k, \mathbf{z}_1, \cdots, \mathbf{z}_m\}$ of V.

Do some work (next two slides) to argue $\{\mathbf{x}_1, \cdots, \mathbf{x}_k, \mathbf{y}_1, \cdots, \mathbf{y}_\ell, \mathbf{z}_1, \cdots, \mathbf{z}_m\}$ is a basis of $U + V$. So

$$\begin{aligned}\dim(U) + \dim(V) &= (k + \ell) + (k + m) = (k + \ell + m) + k \\ &= \dim(U + V) + \dim(U \cap V).\end{aligned}$$

(Span) Let $\mathbf{w} \in U + V$. Then $\mathbf{w} = \mathbf{u} + \mathbf{v}$ for some $\mathbf{u} \in U$ and $\mathbf{v} \in V$.
Since $\mathbf{u} \in U$ and $\{\mathbf{x}_1, \cdots, \mathbf{x}_k, \mathbf{y}_1, \cdots, \mathbf{y}_\ell\}$ is a basis of U, we have
$\mathbf{u} = a_1\mathbf{x}_1 + \cdots + a_k\mathbf{x}_k + a_{k+1}\mathbf{y}_1 + \cdots + a_{k+\ell}\mathbf{y}_\ell$ for some a_i's.
Since $\mathbf{v} \in V$ and $\{\mathbf{x}_1, \cdots, \mathbf{x}_k, \mathbf{z}_1, \cdots, \mathbf{z}_m\}$ is a basis of V, we have
$\mathbf{v} = b_1\mathbf{x}_1 + \cdots + b_k\mathbf{x}_k + b_{k+1}\mathbf{z}_1 + \cdots + b_{k+m}\mathbf{z}_m$ for some b_i's.
So,

$$
\begin{aligned}
\mathbf{w} =& \mathbf{u} + \mathbf{v} \\
=& a_1\mathbf{x}_1 + \cdots + a_k\mathbf{x}_k + a_{k+1}\mathbf{y}_1 + \cdots + a_{k+\ell}\mathbf{y}_\ell \\
&+ b_1\mathbf{x}_1 + \cdots + b_k\mathbf{x}_k + b_{k+1}\mathbf{z}_1 + \cdots + b_{k+m}\mathbf{z}_m \\
=& (a_1 + b_1)\mathbf{x}_1 + \cdots + (a_k + b_k)\mathbf{x}_k + a_{k+1}\mathbf{y}_1 + \cdots + a_{k+\ell}\mathbf{y}_\ell \\
&+ b_{k+1}\mathbf{z}_1 + \cdots + b_{k+m}\mathbf{z}_m \\
\in& \text{Span}\{\mathbf{x}_1, \cdots, \mathbf{x}_k, \mathbf{y}_1, \cdots, \mathbf{y}_\ell, \mathbf{z}_1, \cdots, \mathbf{z}_m\}
\end{aligned}
$$

(Linear independence) Let $a_1, \cdots, a_k, b_1, \cdots, b_\ell, c_1, \cdots, c_m$ such that

$$a_1\mathbf{x}_1 + \cdots + a_k\mathbf{x}_k + b_1\mathbf{y}_1 + \cdots + b_\ell\mathbf{y}_\ell + c_1\mathbf{z}_1 + \cdots + c_m\mathbf{z}_m = \mathbf{0}.$$

Then the vector

$$\begin{aligned}\mathbf{v} &= a_1\mathbf{x}_1 + \cdots + a_k\mathbf{x}_k + b_1\mathbf{y}_1 + \cdots + b_\ell\mathbf{y}_\ell && --(1)\\ &= -c_1\mathbf{z}_1 - \cdots - c_m\mathbf{z}_m && --(2)\end{aligned}$$

is in both U (by (1)) and in V (by (2)), and so $\mathbf{v} \in U \cap V$ and $\mathbf{v} = d_1\mathbf{x}_1 + \cdots + d_k\mathbf{x}_k$. So,

$$\begin{cases} -c_1\mathbf{z}_1 - \cdots - c_m\mathbf{z}_m = d_1\mathbf{x}_1 + \cdots + d_k\mathbf{x}_k & --(3)\\ a_1\mathbf{x}_1 + \cdots + a_k\mathbf{x}_k + b_1\mathbf{y}_1 + \cdots + b_\ell\mathbf{y}_\ell = d_1\mathbf{x}_1 + \cdots + d_k\mathbf{x}_k & --(4)\end{cases}$$

Since $\{\mathbf{x}_1, \cdots, \mathbf{x}_k, \mathbf{z}_1, \cdots, \mathbf{z}_m\}$ is a basis, it is a linearly independent set, and so (3) implies $d_1 = \cdots = d_k = -c_1 = \cdots = -c_m = 0$.
Since $\{\mathbf{x}_1, \cdots, \mathbf{x}_k, \mathbf{y}_1, \cdots, \mathbf{y}_\ell\}$ is a basis, it is a linearly independent set, and so (4) implies $d_1 - a_1 = \cdots = d_k - a_k = -b_1 = \cdots = -b_\ell = 0$.
Since $d_1 = \cdots = d_k = 0$ and $d_1 - a_1 = \cdots = d_k - a_k = 0$, we have $a_1 = \cdots = a_k = 0$.